

Menu-Driven Pyramid / Diamond Pattern Program

C Programming Lab

Aim

To write a menu-driven C program that uses a switch statement and nested loops to print the selected pyramid/diamond pattern. Each pattern is implemented using a separate parameterless function with no return value.

Problem Statement

Display a menu and use a switch statement to print one of the following patterns selected by the user. Use nested loops without arrays. For each activity, prepare the algorithm and test-case table, then write, compile, execute, and test the C program.

1. Full pyramid

Input the number of rows n and print a full pyramid using asterisks.

2. Inverted pyramid

Input the number of rows n and print an inverted pyramid using asterisks.

3. Diamond pattern

Input the number of rows n for the upper half and print a diamond using asterisks. The middle row contains $(2n - 1)$ asterisks and the complete diamond contains $(2n - 1)$ rows.

4. Palindromic number pyramid

Input the number of rows n . In each row, print natural numbers in increasing order from 1 to the row number and then in decreasing order back to 1.

5. Repeated-number pyramid

Input the number of rows n . In each row, print the row number $(2n - 1)$ times, where n denotes the current row number.

Algorithm

1. Start the program.
2. Display the menu containing the five pattern choices.
3. Read the user's choice.
4. Use a switch statement to call the corresponding parameterless function.
5. Inside the selected function, input the required number of rows n .
6. Use nested loops to print spaces, asterisks, or numbers according to the selected pattern.

7. If the choice is not between 1 and 5, display an invalid-choice message.
8. Stop the program.

C Program

```
#include <stdio.h>

void fullPyramid();
void invertedPyramid();
void diamondPattern();
void palindromicPyramid();
void repeatedNumberPyramid();

int main()
{
    int choice;

    printf("1. Full Pyramid\n");
    printf("2. Inverted Pyramid\n");
    printf("3. Diamond Pattern\n");
    printf("4. Palindromic Number Pyramid\n");
    printf("5. Repeated-Number Pyramid\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1: fullPyramid(); break;
        case 2: invertedPyramid(); break;
        case 3: diamondPattern(); break;
        case 4: palindromicPyramid(); break;
        case 5: repeatedNumberPyramid(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

void fullPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
}

void invertedPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
```

```

scanf("%d", &n);
for (i = n; i >= 1; i--)
{
    for (j = 1; j <= n - i; j++) printf(" ");
    for (j = 1; j <= 2 * i - 1; j++) printf("* ");
    printf("\n");
}
}

```

```

void diamondPattern()
{
    int n, i, j;
    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
    for (i = n - 1; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
}

```

```

void palindromicPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        for (j = i - 1; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

```

```

void repeatedNumberPyramid()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("%d ", i);
        printf("\n");
    }
}

```

Test Cases

Choice	Pattern	Input n	Expected Result
1	Full pyramid	4	4-row full pyramid of asterisks
2	Inverted pyramid	4	4-row inverted pyramid of asterisks
3	Diamond pattern	4	7-row diamond; middle row has 7 asterisks
4	Palindromic number pyramid	4	Rows increase then decrease symmetrically
5	Repeated-number pyramid	4	Row i contains the number i exactly $(2i - 1)$ times

Result

The menu-driven C program was written using a switch statement, separate parameterless functions, and nested loops. The selected pyramid/diamond pattern is displayed according to the user's choice and number of rows.

Execution Screenshots

Compilation of the C program (hdco27.c)

```
(kali@kali)-[~/Desktop/apple]
$ gcc hdco27.c -o hdco27

(kali@kali)-[~/Desktop/apple]
$
```

Output 1: Full Pyramid (n = 4)

```

(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
Enter your choice: 1
Enter number of rows: 4
  *
 * * *
* * * * *
* * * * * * *

```

Output 2: Inverted Pyramid (n = 4)

```

(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
Enter your choice: 2
Enter number of rows: 4
* * * * *
 * * * *
  * * *
   *

```

Output 3: Diamond Pattern (n = 4)

```

(kali@kali)-[~/Desktop/apple]
$ ./hdoc27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
Enter your choice: 3
Enter number of rows for upper half: 4

  *
 * * *
* * * * *
* * * * * *
 * * * *
  * * *
   *

```

Output 4: Palindromic Number Pyramid (n = 4)

```

(kali@kali)-[~/Desktop/apple]
$ ./hdoc27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
Enter your choice: 4
Enter number of rows: 4

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1

```

Output 5: Repeated-Number Pyramid (n = 4)

```

(kali@kali)-[~/Desktop/apple]
$ ./hdoc27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
Enter your choice: 5
Enter number of rows: 4

1
2 2 2
3 3 3 3 3
4 4 4 4 4 4 4

```